

Improving Software-Defined Networks Performance using Machine Learning

(Project-Based)

Amal Majed ALOtaibi

461215785

Supervised by: Dr. Lamia Amor Benabderrahman

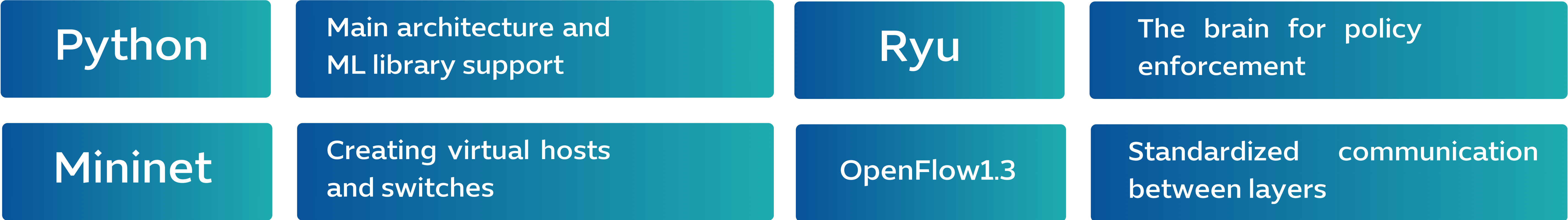
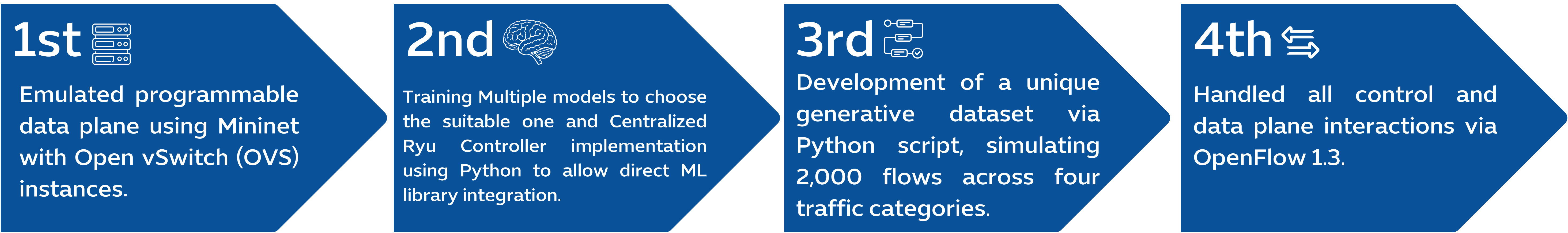
College of Computer, Computer Science Department



01 Introduction

Traditional Software-Defined Networking (SDN) faces significant stability challenges when managing dynamic traffic loads and encrypted applications, which often leads to degraded Quality of Service (QoS) By utilizing a Mininet-emulated environment With using Random Forest in the Ryu controller, the system automates real-time traffic classification and dynamic policy enforcement to enhance critical performance metrics such as throughput, latency, and jitter.

02 Methodology



03 Experiments

The experiment evolved from training ML models on static generative datasets to testing in a dynamic 4-host Mininet topology. A 10Mbps bottleneck was created to challenge the network with concurrent flows: critical VLC video and a 100Mbps UDP flood.

Metrics	Inference Latency	AVG Jitter (MS)	Throughput	QOS Decision
Base (without ML)	0.8	0.191	7.6	Hardcoded
RF	0.4	0.142	10.5	Driven AI (Dynamic)

The RF model improves the results by dynamically elevating critical flows to Priority 800, increasing throughput by 38% from 7.6 to 10.5 Mbps. This intelligent queuing reduces average jitter by 25% from 0.191 to 0.142 ms and cuts latency by 50% compared to the static baseline.

04 Conclusion

The Random Forest (RF) model serves as the primary intelligence engine, enabling the Ryu controller to autonomously classify real-time traffic into distinct service categories: Video, Image, Web, and Text. By analyzing dynamic flow behaviors, the RF-SDN framework moves beyond static rule-based switching to intelligent, application-aware networking. Current experimental results demonstrate a significant performance delta, where the ML-driven prioritization (Priority 800) effectively mitigates congestion compared to the standard Baseline. To finalize this evaluation, we are integrating multi-flow UDP traffic analysis and performance curves to provide a definitive quantitative proof of the model's impact on network stability and Quality of Service (QoS).

